Job Insights 2024 Analysis

Practice Questions & Solutions

Q1. **Retrieve the top 10 highest-paid Data Analyst jobs. Exclude records with NULL salary values. Only include jobs where the location is 'Anywhere'.**

Solution:

```
SELECT
    job_id,
    company_id,
    job_title_short,
    job_title,
    salary_year_avg
FROM job_postings_fact
WHERE job_title_short IN ('Data Analyst')
      AND
      salary_year_avg IS NOT NULL
      AND
      job_location IN ('Anywhere')
ORDER BY 5 DESC
LIMIT 10;
```

Q2. **Retrieve the top 10 highest-paid Data Analyst jobs. Exclude records with NULL salary values. Only include jobs where the location is 'Anywhere'. What skill is required for the post?**

Solution:

```
SELECT
    j.job_id,
    j.company_id,
    j.job_title_short,
    j.job_title,
    j.salary_year_avg,
    s.skills
FROM job_postings_fact j JOIN skill_job_dim sdm ON j.job_id =
sdm.job_id
                         JOIN skill_dim s       ON sdm.skill_id =
s.skill_id
```

```
WHERE j.job_title_short IN ('Data Analyst')
      AND
      j.salary_year_avg IS NOT NULL
      AND
      j.job_location IN ('Anywhere')
ORDER BY 5 DESC
LIMIT 10;
```

Q3. What are the top 5 most demanded skills for the Data Analyst role?

Solution:

```
SELECT
    s.skills,
    COUNT(j.job_id) AS job_count
FROM job_postings_fact j JOIN skill_job_dim sdm ON j.job_id =
sdm.job_id
                          JOIN skill_dim s       ON sdm.skill_id =
s.skill_id
WHERE j.job_title_short IN ('Data Analyst')
GROUP BY 1
ORDER BY 2 DESC
LIMIT 5;
```

Q4. What are the top 10 highest-paying job postings for the 'Data Analyst' role that require SQL skills?

Solution:

```
SELECT
    j.job_id,
    j.company_id,
    j.job_title_short,
    j.job_title,
    j.salary_year_avg,
    s.skills
FROM job_postings_fact j JOIN skill_job_dim sdm ON j.job_id =
sdm.job_id
                         JOIN skill_dim s       ON sdm.skill_id =
s.skill_id
WHERE j.job_title_short IN ('Data Analyst')
```

```
      AND
      j.salary_year_avg IS NOT NULL
      AND
      j.job_location IN ('Anywhere')
      AND
      s.skills ILIKE ('%sql%')
ORDER BY 5 DESC
LIMIT 10;
```

Q5. Most demanded and highest-paying skill for Data Analyst?

Solution:

```
SELECT
    s.skills,
    COUNT(j.job_id) AS demand_count,
    ROUND(AVG(j.salary_year_avg), 2) AS avg_salary,
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY j.salary_year_avg) AS
median_salary
FROM job_postings_fact j JOIN skill_job_dim sdm ON j.job_id =
sdm.job_id
                         JOIN skill_dim s       ON sdm.skill_id =
s.skill_id
WHERE j.job_title_short = 'Data Analyst'
     AND
     j.salary_year_avg IS NOT NULL
GROUP BY 1
ORDER BY 2 DESC
LIMIT 10;
```

Q6. How many Data Analyst job postings were made in each month?

Solution:

```
SELECT
    TO_CHAR(job_posted_date::DATE, 'YYYY-MM') AS year_month,
    TO_CHAR(job_posted_date::DATE, 'Mon'),
    COUNT(job_id) AS total_job_post
FROM job_postings_fact
GROUP BY 1, 2
ORDER BY 1;
```

Q7. Add a new column based on salary. Find all salary_category for Data Analyst.

Solution:

```
SELECT
    job_id,
    company_id,
    job_title_short,
    job_title,
    salary_year_avg,
    CASE
        WHEN salary_year_avg >= 100000 THEN 'Desired'
        WHEN salary_year_avg >= 50000  THEN 'High'
        WHEN salary_year_avg >= 30000  THEN 'Standard'
        ELSE 'Low'
    END AS category
FROM job_postings_fact
WHERE job_title_short IN ('Data Analyst')
      AND
      salary_year_avg IS NOT NULL;
```

Q8. Find the company with the most Data Analyst job openings. Return with company name and total job count.

Solution:

```
SELECT
    c.company_id,
    c.name,
    COUNT(j.job_id) AS job_count
FROM company_dim c LEFT JOIN job_postings_fact j ON c.company_id =
j.company_id
WHERE j.job_title_short IN ('Data Analyst')
GROUP BY 1, 2
ORDER BY 3 DESC;
```